

Modeling of single inductor based Battery Balancing Circuit for Hybrid Electric Vehicles

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Abstract—Li Ion batteries are the proficient energy source for powering Hybrid Electric Vehicles (HEV) due to its smaller size, lighter weight, and improved charge efficiency without memory effect. The battery pack constitutes of a combination of cells in series and parallel configurations. Due to cell impedance mismatch, each cell possesses different potentials which cause different State of Charge (SOC) of cell. Battery balancing is a key issue which aims to maximize the capacity of the battery pack to make all of its energy available for use and also increase the battery's endurance. This paper presents an automatic battery balancing circuit model by using single switch inductor with SOC based logic controller. The performances during static, charging, discharging and mixed conditions are analyzed. The main advantage of the proposed approach is that it has fastest equalisation time due to constant inductor current which is controlled by Proportional Integral (PI) controller and SOC-based logic controller for controlling the switching action.

Keywords—Battery pack, Power MOSFET switch, PI controller, SOC-based logic controller.

I. Introduction

The basic requirements of an automotive class of lithium ion battery systems are:

- High Energy without conceding power performance: Higher charge and discharge rates for better performance and efficiency
- Higher Usable Energy: Wide SOC range to enable greater battery utilization
- Extended Cycle Life in a compact package: Long battery life with 100% Depth of Discharge (DOD) cycles
- Excellent Safety with superior abuse tolerance

The battery packs in HEV and Electric Vehicles are exposed to the most critical environment as they are always subjected to dynamic conditions and transients due to the constant charging and discharging phenomenon occurring due to acceleration, deceleration and regenerative braking which cause mismatch in the cell parameters. Also, the difference in chemical and electrical characteristics during manufacturing of cell results in a mismatch in the cell parameter i.e. voltage and SOC.

A battery pack consists of individual cells configured in series and parallel combinations. Since, the individual cells in a battery hold different capacities and may be at different levels of SOC, the energy that can be charged and discharged from the battery pack is limited to the cell with the least capacity and it can be easily overcharged or over-

discharged while cells with higher capacity may undergo only partial cycles. For the higher capacity cells to undergo full charge/discharge cycles without overcharging or over discharging any other cell in the battery pack battery balancing is required. Battery balancing can be achieved either by equalising the voltage or SOC. To extend the battery life Cell balancing based on SOC is proposed in this paper where, battery balancing is done by transferring energy between individual cells, until the SOC of the cell with the lowest capacity is equal to the battery's SOC.

II. Battery Balancing techniques

SOC imbalance, total capacity imbalance and impedance imbalance causes voltage differences amongst cells in a battery pack. SOC imbalance can be mitigated by bypassing certain amount of charge, total capacity imbalance is taken into account for bypass charge calculation considering impedance imbalance as distortion. Battery Management System (BMS) is designed to maximize the life of the battery pack and the major subsystems of BMS are battery balancing, temperature monitoring and charging circuitry ^[1]. Numerous charge-balancing algorithms and circuits have been developed ^{[2]-[4]}. Battery balancing can be classified as active and passive. The passive method is of discharge type because it dissipates energy by resistor during balancing. In active method, the charge is extracted from the cell with the highest SOC in the battery pack by using storage element i.e. inductor and capacitor and the extracted charge is stored to other cells. So, the active method can be of charge type, discharge type or a composite charge-discharge type. Also, the cell balancing for fewer numbers of cells, automatic or selective equalization based on a multiwinding transformer ^[5], intelligent cell balance using dc-dc converter ^[6], equalization using bidirectional dc-dc converter ^[7] are proposed. In practical scenarios to obtain a higher dc voltages hundreds of batteries are stacked together and in such case it is difficult to implement a multiwinding transformer or control a large number of bidirectional dc-dc converters.

Several battery balancing topologies have been reviewed and simulated using MATLAB/Simulink ^{[8]-[9]}. The broad comparison of balancing techniques provides guidance to select the optimal solution for the given application. The advantages and disadvantages of various cell balancing circuits are tabulated below in Table I and automatic battery balancing circuit model by using single inductor with SOC based switch controller as shown in Fig.1.

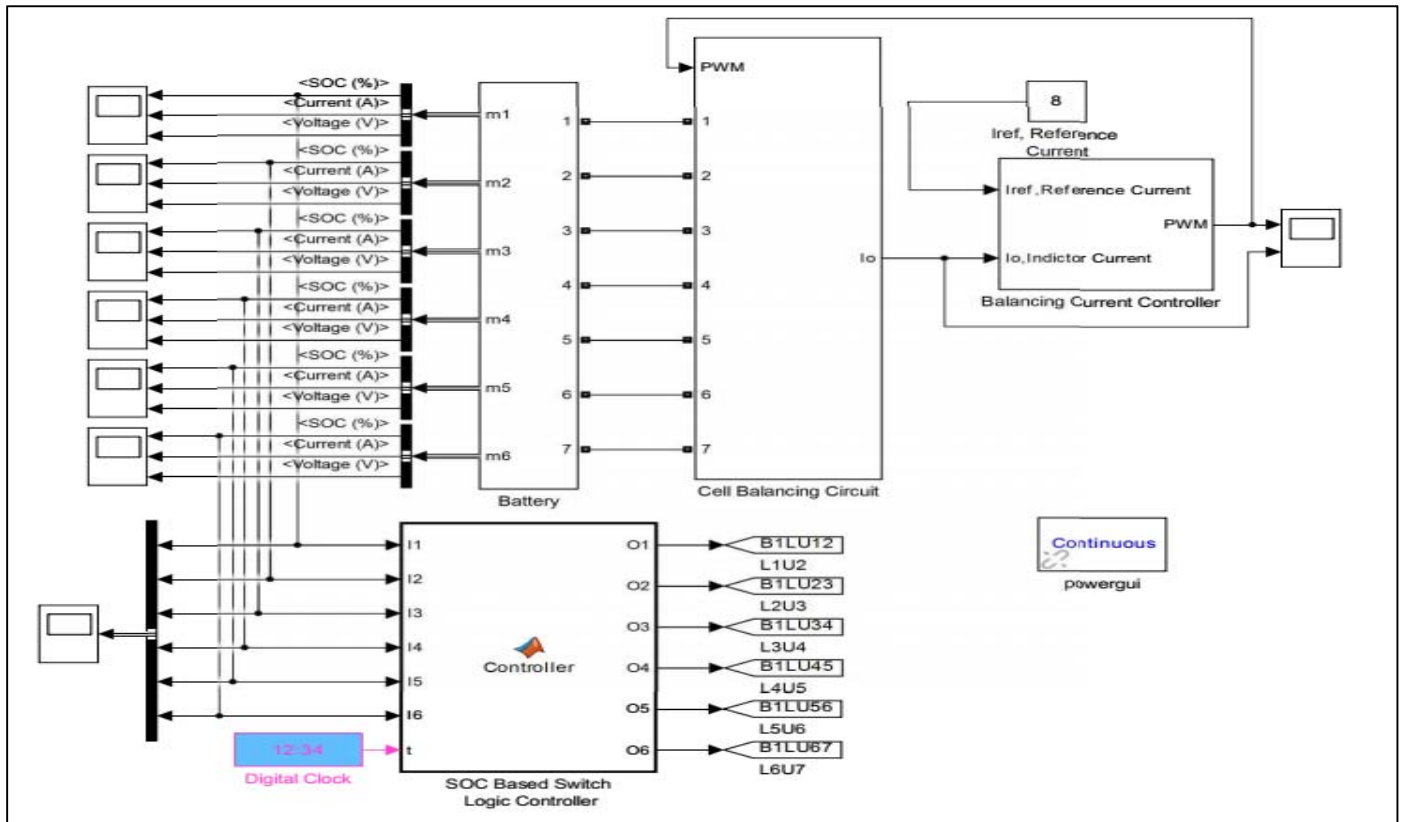


Fig. 1. Automatic Battery Balancing circuit model

TABLE I. ADVANTAGEES AND DISADVANTAGES OF VARIOUS CELL BALANCING CIRCUIT

Scheme	Advantage	Disadvantage
Fixed Resistor	Cheap and easy to implement	High energy loss
Switched capacitor	Simple control	Low equalisation rate, large number of switches
Modularized switched capacitor	Low voltage and current stress	Complex control, satisfactory equalisation speed
Single Inductor	Fast equalisation speed, good efficiency	Complex control, filtering capacitors needed for high switching frequency
Multi inductor	Fast equalisation speed, good efficiency	Less complex control. Accurate voltage sensing needed. Filtering capacitors are needed for high switching current.
Single Windings Transformer	Fast equalization speed, Low magnetic losses.	Expensive , bulkier and Complex control, to add one or more cells the core must be changed
Buck-Boost Converter	Good equalization speed, Easy for modular design.	High cost, Intelligent control needed.
Fly back converter	Suitable for modularized system	Voltage sensing needed, satisfactory equalisation needed
Quasi Resonant Converter	High efficiency	High cost, complex and intelligent implementation needed

III. Proposed Balancing Circuit Model

The proposed active cell balancing topology uses inductors to transfer energy from a cell or group of cells. In this model the SOC of the individual cells is sensed using SOC based switch controller and corresponding switches of the balancing circuit is activated or deactivated for balancing the cells of the battery pack. A constant balancing current is maintained using PI based PWM Controller. The automatic battery balancing circuit model consists of battery pack, balancing circuit, balancing current controller and SOC based switch controller.

A. Battery Pack

The cell potential ranges from 3.0 to 4.4V with Li ion cell chemistry. During discharging it is essential to maintain the minimum cell voltage of 3.0 V to prevent deep discharge which can create a catastrophic effect on the cell recovery and also may lead to short circuit. During charging it is required to maintain a maximum cell voltage of 4.3V to prevent overheating and damage of the cell.

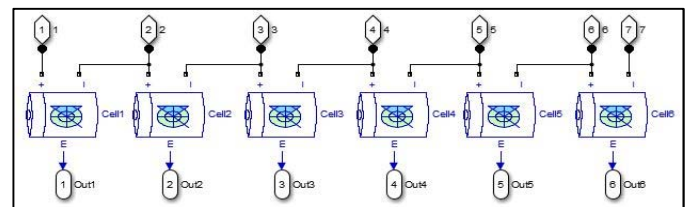


Fig. 2. Li-ion Battery Pack

In this model, to simulate 24V, 75Ah capacity battery pack six Li-ion cells with 3.7V nominal voltage and capacity 75Ah^[10] is connected in series.

B. Single Inductor based balancing System

The single inductor based balancing system, utilizes one inductor for transferring energy between the whole pack as marked as blue in shown in Fig 3. The control system senses the SOC of the cells and selects the cell with lowest SOC for energy transfer through activation of the corresponding switches.

The circuit has battery pack which has n number of cell in series. The balanced circuit has two group of switching device each group have n+1 Power MOSFET with parasitic diode in inverse parallel which acts as a switch which is connected with single inductor. Also, a diode is in series with each Power MOSFET to prevent the cell from being shorted during balancing. Hence, Total 2n+2 number of power MOSFET and diode are used but during the balancing any of four switches is in operation.

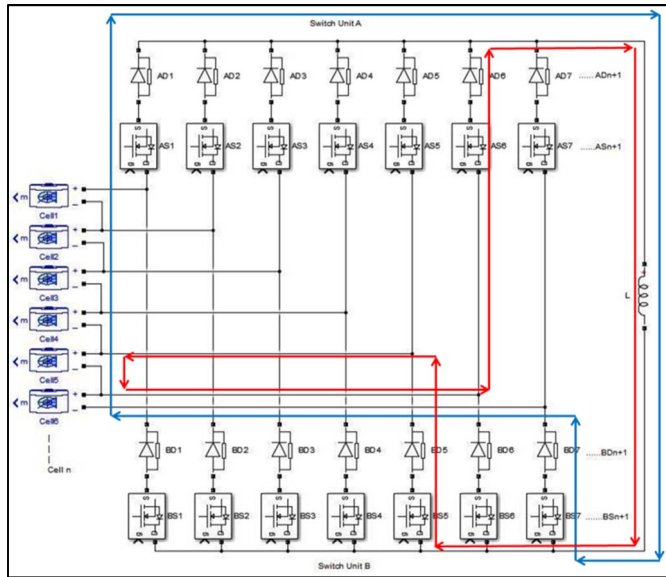


Fig. 3. Single inductor based cell balancing circuit

C. Working Principle

In this battery balancing circuit, the inductance is used to store energy from battery pack through the two group of switching device and transfer the energy from the whole stored energy of inductor to the cell which has lowest voltage or SOC. E.g. the battery pack has string of six cells, and assuming that the voltage or SOC of cell 5 is lowest. So, the switch AS1 and BS7 only are controlled to turn on or turn off state to store or release the energy into inductor, while AS6 and BS5 are being in turn on state to store the energy in the cell 5. Hence when AS1 and BS7 are turned on then inductor store energy from battery pack and when they are turned off inductor releases energy into cell 5. The balancing scheme depends on lowest SOC cell. The Table II gives the switching action of the balancing circuit.

TABLE III. SWITCHING ACTION OF BALANCING CIRCUIT

Sl. No.	Lowest SOC Cell	Switch ON
1	Cell1	AS2 and BS1
2	Cell2	AS3 and BS2
3	Cell3	AS4 and BS3
4	Cell4	AS5 and BS4
5	Cell5	AS6 and BS5
6	Cell6	AS7 and BS6

D. Constant balancing current control Mode

There are two control modes for balancing current i.e. discontinuous current mode and continuous current mode.

a) Discontinuous Current Mode

In this mode the working waveform are shown in Fig. 4^[11]. Assume that, the switching cycle of AS1 and BS7 is T which is equal to t₃. So the duty ratio, α is

$$\alpha = \frac{t_1}{t_3} = \frac{t_1}{T} \quad (1)$$

Where

$$t_2 \leq T$$

Let, the total voltage of the battery pack is V and V₅ is the voltage of cell5, the maximum balanced current is fixed to I_{max}. If i_{cell5} > 0 means the battery pack is discharging and i_{cell5} < 0 means cell5 is being charging.

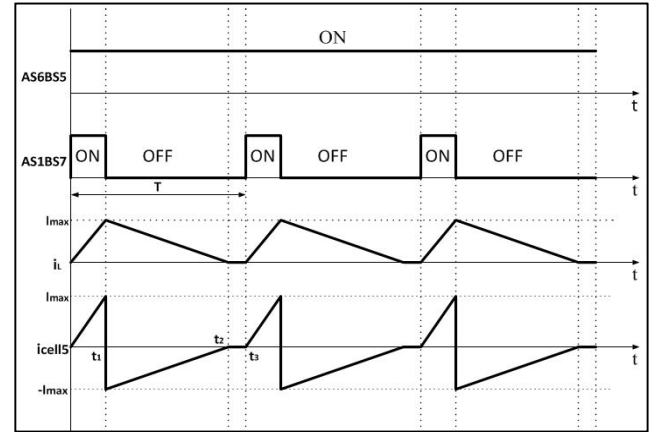


Fig. 4. Discontinuous Current mode

During ON period of switching cycle i.e. 0 to t₁, AS1 and BS7 are turned on. The current i_L of the inductor is rising faster, and at the end of t₁, i_L reaches the maximum value I_{max}, i_L is described as

$$i_L(t) = \frac{V}{L} t \quad (2)$$

Where

$$i_L(t_1) \leq I_{max}$$

During OFF period of switching cycle i.e. t₁ to t₂, AS1 and BS7 are turned off, AS6 and BS5 are turned on, the inductor current i_L is dropping slowly, and at the time t₂ it is equal to zero. i_L is described in (3)

$$i_L(t) = I_{max} - \frac{V_5}{L} (t - t_1) \quad (3)$$

$$i_L(t_2) \leq 0$$

Therefore, the parameter can be deduced from above equation (1), (2) and (3) are $T = 0.005$ sec, $\alpha = 21\%$ and $L = 2\text{mH}$ for the maximum inductor current, $I_{\text{max}} = 8\text{A}$. Hence, the discontinuous current mode simulation results are shown in Fig.5.

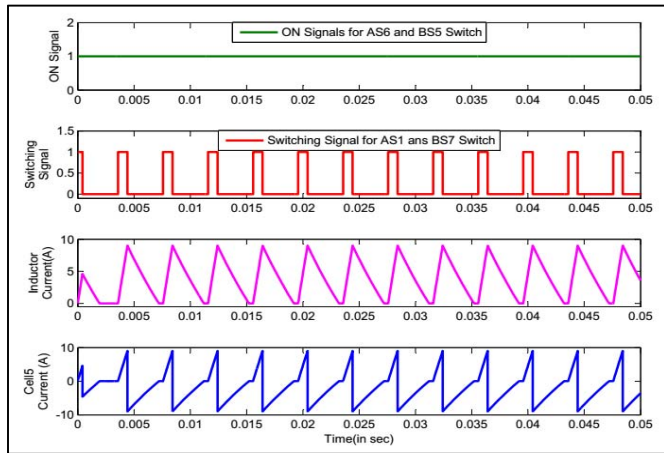


Fig. 5. Discontinuous Current mode Simulation

It is observed that in discontinuous mode, the frequency of the PWM pulse balancing current has zero cross over within time period. So the equalization time is more in discontinuous mode.

b) Continuous Current Mode

In this mode the working waveform are shown in Fig. 6^[11].

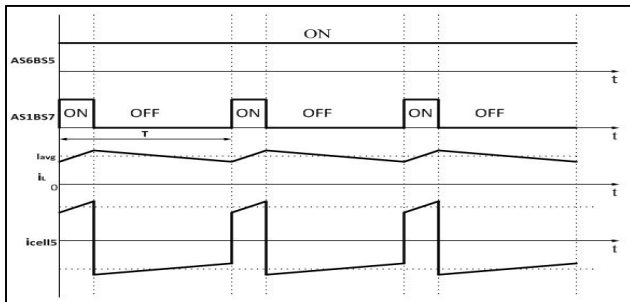


Fig. 6. Continuous current mode

Switching of AS1 and BS7 is controlled using a PWM for storing the charge in the inductor. To maintain a constant inductor current PI controller is introduced as shown in Fig 7.

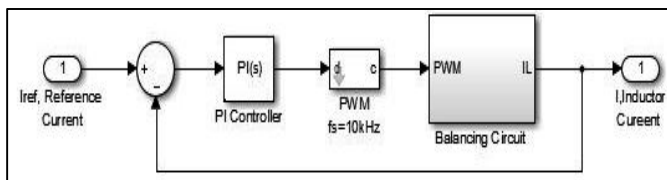


Fig. 7. Constant current mode

In this circuit, the inductor or balancing current, I_L is to be maintained 8A which is developed by inductor current

feedback with PI controller followed by PWM with frequency 10 kHz.

E. SOC based switch Controller

It contains MATLAB based program with respect to lowest SOC cell in which input are SOC of battery pack cell and it generate the output according to SOC comparisons that enable and disable the MOSFET switch ON or OFF. This program automatically enables and disables the switch according to cell SOC comparisons. The flow chart for SOC based switch controller algorithm is shown in Fig 8.

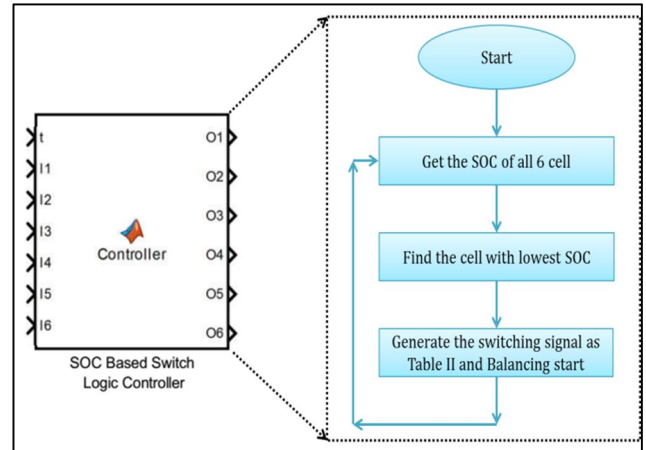


Fig. 8. SOC based Switch Controller

iv. Simulation Result

The battery pack consists of six Li-ion cells in series and cell model is taken from Matlab-Simulink. The nominal voltage of Li-ion cell is 3.7V and nominal capacity is 75Ah.

TABLE IV. INITIAL SOC OF BATTERY PACK

Cell No.	Assumed Initial SOC
Cell1	60.06
Cell2	60.05
Cell3	60.04
Cell4	60.03
Cell5	60.01
Cell6	60.02

The lowest SOC cell is detected by SOC based switch controller and its output drives switch of balancing circuit.

The battery balancing circuit is simulated for four different modes: a) Static mode, b) Charging mode c) Discharging mode and d) Mixed mode.

A. Static Mode

When the battery pack is connected only with the balancing circuit switches AS1 and BS7 are turned ON and energy from cells with higher SOC are transferred to the cells with lower SOC till all the cells are maintained in the same level of Charge. It can be inferred from Fig. 9 that the SOC of

battery pack converges to a constant level of charge in 70 secs for the assumed initial SOC given in Table III.

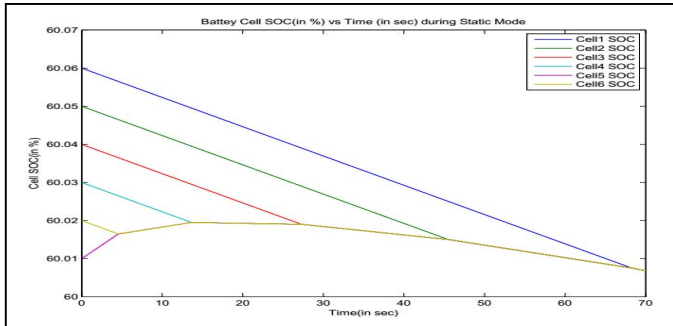


Fig. 9. Cell Balancing during static mode

B. Charging Mode

In this mode, the battery pack is connected to constant current source for 1C rate with cell balancing circuit to charge the battery pack with cell balancing. If the SOC of any individual cell is greater than the maximum threshold it is isolated from the charging circuit to prevent overcharging. It is observed that SOC of each cell is converged to a steadily increasing value during battery charging as shown in Fig 10.

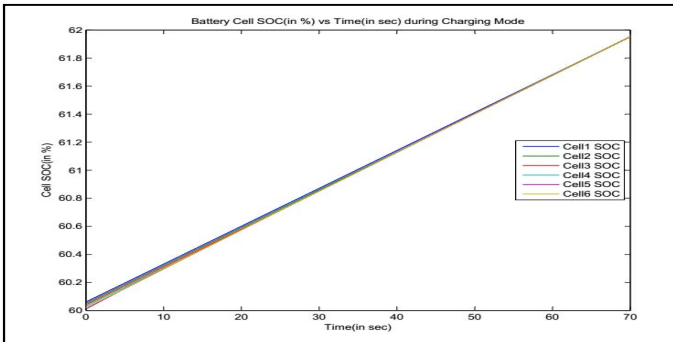


Fig. 10. Cell Balancing during charging

C. Discharging Mode

In this state, the battery pack is connected to resistive load with cell balancing circuit to discharge the battery pack for 1C rate with cell balancing. If the SOC of the individual cell is less than the minimum threshold switching sequence is altered to prevent deep discharge. The SOC response of cell of battery pack during discharging is shown in Fig. 11.

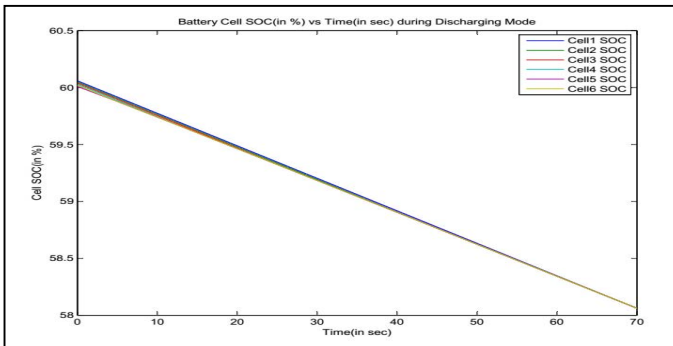


Fig. 11. Cell Balancing during discharging

D. Mixed Mode

In HEV, the battery pack has mixed state i.e. combination of static, charging and discharging states. To analyze this situation, a current signal is created using signal builder with combination of static, charging and discharging state with different C-rate as shown in Fig 12.

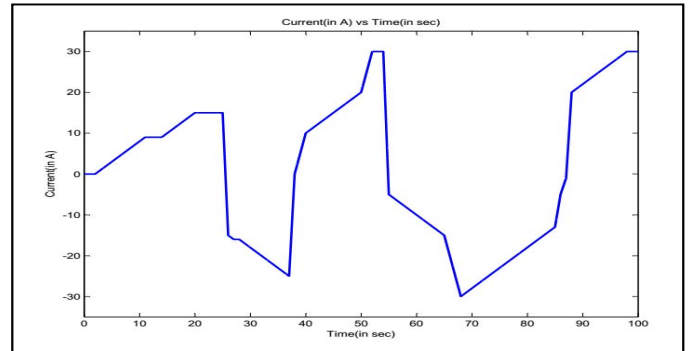


Fig. 12. Mixed Current Signal

Hence, the response of cell of battery pack is shown in Fig.13.

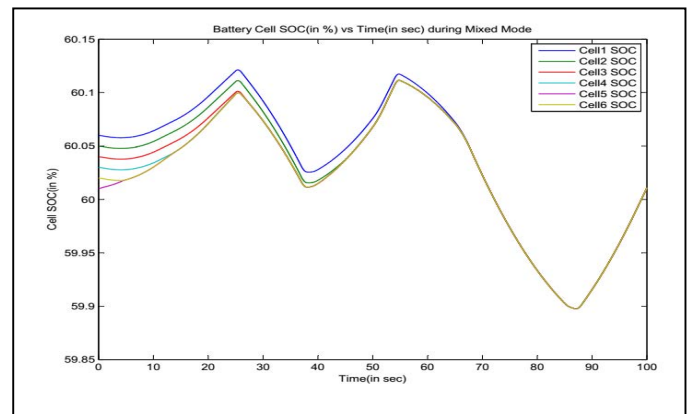


Fig. 13. Cell Balancing in Mixed Mode

v. Conclusion

This paper proposed an automatic battery balancing circuit with SOC based switch controller that balances the SOC of cell with PI controller based constant balancing current. It is observed that once the cell of battery pack is balanced then all cells have same SOC during charging or discharging or steady state.

vi. Future Work

It is proposed to develop temperature monitor circuit, charging circuit with balancing circuit that incorporate an Intelligent Battery Management system. Also, the Cooling system issue for battery pack will be analyzed by FEM tool.

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